

SIH26137 — Concept Prototype

Quantum-Inspired Intelligent Traffic Route Optimization

Create a professional 6-slide technical prototype presentation for **SIH26137 — Quantum-Inspired Intelligent Traffic Route Optimization in Transportation Systems Using Metaheuristic Optimization**.

The presentation should communicate one core idea:

Use a quantum-inspired metaheuristic optimization engine to dynamically generate high-quality vehicle routes under changing traffic conditions, and validate it against classical algorithms.

Keep the presentation technical, visual, and concise. Avoid unnecessary business features.

Slide 1 — Problem & Motivation

Title

Dynamic Transportation Routing is a Large-Scale Optimization Problem

Key points

- Urban transportation networks contain **many vehicles, destinations, roads, and constraints**.
- Traffic conditions continuously change, making previously optimal routes obsolete.
- Vehicle Routing Problem (VRP) is **NP-hard**; exhaustive search becomes impractical as problem size grows.
- Classical shortest-path algorithms solve individual routes but do not efficiently solve the complete **multi-vehicle routing problem**.

Visual

Show: **Road Network → Multiple Vehicles → Multiple Destinations → Traffic Congestion → Complex Route Decisions**

Bottom message

We need fast, scalable optimization—not just shortest-path calculation.

Slide 2 — Our Concept Prototype

Title

From Traffic Data to Optimized Routes

Show the prototype as a simple pipeline:

Road Network ↓ **Graph Construction** ↓ **Traffic Model** ↓ **VRP + Constraints** ↓ **QPSO Optimization Engine** ↓ **Optimized Vehicle Routes** ↓ **Map + Performance Analytics**

Prototype scenario

Use a simulated urban environment:

- 20–50 road/network nodes
- 10–30 delivery/customer locations
- 3–5 vehicles
- Different vehicle capacities
- Simulated traffic conditions
- Depot/start/end constraints
- Optional time windows

Key message

The prototype demonstrates the complete optimization loop before building the full platform.

Slide 3 — How the Optimization Works

Title

QPSO as the Optimization Engine

Explain at a high level:

Problem representation

Transportation network:

$G = (V, E)$

where:

- V = intersections/locations
- E = roads

- Edge weight = travel time / distance / congestion cost

Objective

Minimize:

$$\text{Total Cost} = \alpha \times \text{Travel Time} + \beta \times \text{Distance} + \gamma \times \text{Congestion}$$

subject to:

- Vehicle capacity
- Maximum route duration
- Depot constraints
- Time windows
- Vehicle availability

Optimization idea

Each candidate solution represents a possible set of vehicle routes.

Generate solutions → Evaluate fitness → Search promising regions → Update solutions → Repeat → Best feasible route

Important note

QPSO is quantum-inspired and runs on classical computing hardware.

Slide 4 — Dynamic Traffic & Re-Optimization

Title

The Route Adapts When Traffic Changes

Show a before/after scenario.

Scenario

Initial condition

- Road A has low traffic
- Optimizer selects Route A

↓

Traffic incident

- Road A becomes highly congested

↓

Updated edge weights

- Travel time/congestion cost increases

↓

Re-optimization

- QPSO evaluates alternative routes

↓

New route

- Vehicle is redirected through Route B

Visual

Use a city map with:

- Vehicle icons
- Destination points
- Original route
- Congested road
- New optimized route

Key message

The system optimizes against the current transportation state, not a static map.

Slide 5 — Validation & Benchmarking

Title

We Don't Assume QPSO is Better — We Measure It

Compare the same routing problem using:

Algorithm	Solution Quality	Runtime	Convergence
Greedy / Nearest Neighbour	Baseline	Low	Fast
Genetic Algorithm	—	—	—
Classical PSO	—	—	—
QPSO	—	—	—
Exact Solver*	Optimal	High	—

Metrics

- Total travel cost
- Total travel time
- Distance
- Runtime
- Convergence speed
- Optimality gap
- Constraint violations
- Scalability

Experimental principle

Same dataset + same constraints + same objective + multiple algorithms

*Use exact optimization primarily on small instances where computing the optimum is practical.

Key message

The prototype is also an experimental framework for scientifically evaluating QPSO.

Slide 6 — Prototype Outcome & Next Phase

Title

What This Prototype Must Prove

The prototype should answer four questions:

1. Can we model transportation as a dynamic optimization problem?

Graph + traffic + vehicles + constraints

2. Can QPSO produce feasible, high-quality routes?

Route quality + constraint satisfaction

3. Does dynamic traffic trigger useful re-optimization?

Traffic change → new route

4. How does QPSO compare with classical approaches?

Quality + runtime + convergence + scalability

Prototype Output

Interactive Route Map

+ Optimized Vehicle Routes

+ Traffic Re-routing Demonstration

+ Algorithm Benchmark Dashboard

Next phase

Prototype Validation → PRD → Mathematical Specification → Algorithm Engineering → Full MVP

Final statement

Our goal is not simply to find a route.

Our goal is to build and validate an optimization engine for dynamic, large-scale transportation routing.